SSHVisual: Username-Centric Visual Analysis of SSH Attacks

Shreeman Gautam
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

Robert Ricci
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

***Abstract*—Network operators use visualizations to detect attacks on their systems and uncover hidden patterns. Operators use these findings to block attackers and expand the knowledge base of known threats. Iteratively detecting anomalies and relationships in data can be tedious at best and impractical at worst without visualization. Much prior work in network visualization analyzed the IP and ASN dimensions of SSH logins with little focus on the username, and detected anomalies only when baseline events were available. As a result, little attention has been paid to whether attackers that differ in timing, source, and username belong to the same campaign. We present SSHVisual, a visualization system evaluated on 206.6 million login attacks against our production SSH systems over 69 days. SSHVisual analyzes the IPs, ASNs, and usernames of SSH attacks to support campaign detection. It complements automated detection, letting operators find anomalous events without a normal baseline. Its similarity search groups attackers by behavior or by the usernames they share, and explains each match. As in prior tools, operators can tag findings and share them. Case studies suggest it can help operators surface hidden patterns and find attackers to block. In one study, six IPs in four countries tried the same 77 usernames, including 16 uncommon ones, pointing to a single campaign that volume charts could show as unrelated bursts. In another, similarity search surfaced low-volume IPs working through large, overlapping username lists; none ranked in the top 10 by volume. These results suggest that username-centric, operator-driven exploration can reveal covert, coordinated activity that volume-based, IP-centric analyses may miss.**

***Index Terms*—SSH, brute-force attacks, attack campaigns, network security visualization, visual analytics, similarity search**

I. Introduction

Computer systems employ SSH [1] to let users connect securely, run commands, and perform remote administration. Many users gain access with a username and password (password-based authentication). Attackers try to brute-force [2] their way into SSH systems. They guess username–password combinations to exploit poorly secured systems, which they then use to build botnets [2] for further attacks [3], [4] or to mine cryptocurrency with stolen resources [5], [6]. Given the prevalence of SSH attacks in the IPv4 space [2], we designed SSHVisual and evaluated it on login attacks from our production SSH systems. SSHVisual serves two kinds of network operators: network administrators, who handle daily SSH traffic and need to quickly identify and block attackers; and security analysts, who study traffic over longer periods, look for hidden patterns, and share findings. For example, in a single period, an administrator can block the highest-volume attacker, while an analyst can iteratively surface low-volume attackers that attempt the same usernames and group them as a campaign.

Surveys of network visualization [7]–[11] map an extensive body of network visualization work, spanning management, security, anomalies, routing, and DDoS detection. We treat the username, alongside the IP and ASN, as part of an attacker's identity, and make three contributions. First, we derive tasks and requirements for the two operators (Section III). Second, we present SSHVisual, which pairs a ranked discovery table and coordinated line charts with similarity search over usernames and IPs, and with shared tagging, as in [12] (Section IV). Third, two case studies (Section V) illustrate how SSHVisual serves each operator. For analysts, username-based similarity can identify campaigns that volume-based views may split apart: IPs that try the same usernames likely belong to one campaign, even when they span countries and ASNs and peak on different days. For administrators, behavioral similarity search can find attackers that volume rankings may hide. Operational monitoring can focus on the largest flows [13], and threshold detectors can be evaded by spread-out attackers [14], [15], with Gu et al. [16] showing that shared behavior can reveal host relationships. We make this actionable: starting from an IP the administrator wants blocked, our search surfaces low-volume IPs that behave alike but miss the top 10 rankings, and explains each match, so the administrator can see which IPs to block and why.

Since network traffic may not offer a stable, attack-free baseline [17], baseline-driven detectors may miss activity that does not deviate from that baseline. Operators, by contrast, may recognize coordination and similarity across sessions visually, without specifying normality. Our contributions build on this ability and are designed to complement baseline-driven detection. As operators explore, they can label and share findings so colleagues avoid repeat investigations, and those findings can in turn inform automated detection. The rest of the paper is organized as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation, and Section VI is Conclusion.

II. Related Work

Visualization tools [12], [18]–[26] make network traffic and security events navigable, most being traffic-oriented [18], [19], [21], [23]–[26]. Mansmann et al. [20] combine intrusion-detection events with NetFlow data, Chen et al. [22] combine IPS and host logs, and Goodall and Sowul [12] show multi-dimensional cyber-defense data. Livnat et al. [27] visualize IDS alerts, Humphries et al. [28] visualize heterogeneous security logs, and Yu et al. [29] map malicious activity from firewall and intrusion logs at the city level. Jeong et al. [30] extract multi-resolution discrete-wavelet features from traffic, project them with principal component analysis, and apply hierarchical clustering across intrusion-detection benchmarks, separating brute-force attacks from normal traffic. Related work on coordinated multiple views [31]–[33] and insight generation [34] informs our design.

A. Username as an Actor

Most SSH credential analysis comes from honeypots, which may see the same attacks as production systems; to our knowledge, that literature has rarely treated the username as an actor. Typically, it is a dimension to count rather than track, whereas in our analysis it serves, alongside the IP and ASN, as an anchoring dimension. Several papers [35]–[41] collect or chart honeypot traffic, with some capturing SSH credentials. Where credentials are charted, such as the SCADA SSH honeypot [38], kippo-graph [39], and the container deployment [40], usernames and passwords appear as static top-N tallies (e.g., kippo-graph's top 10 bars). The WAN honeypot framework [42] is the closest prior system to ours: it computes an IP-to-IP dictionary-overlap score to probe whether six IPs from the same /24 subnet share a wordlist, a manual, one-off prototype of credential-based coordination analysis per IP.

B. Username-Based Similarity

Grouping similar entities is well established [43]–[46]. We do not claim similarity computation as novel; our contribution is the username entity. Prior systems cluster IPs by port behavior [43], IPs by their diurnal traffic profiles [45], IP/port members by their alert and flow timelines [46], or enterprise users by shared hosts [44]. None of these treat the username as an attacker entity. We use the username as an IP signature, grouping IPs by the credentials they try, with uncommon usernames weighted as stronger evidence than ones nearly every attacker tries. Analyses have linked attack activity over time by grouping attacks whose time series rise and fall together [47], [48] and by diurnal profiles [45]. Attackers that share a credential list, however, need not be active together, so our shared-username search ignores when each attacker is active and lets the operator see their time series in a line chart.

C. Limits of Automated Detection

Many anomaly detectors assume a normal baseline, and SSH traffic generally does not have one. Even where a baseline exists, such detection can be fragile. Sommer and Paxson [49] argue that machine-learning anomaly detection is rarely deployed because normal traffic varies widely, errors are costly, and sound evaluation is hard. Axelsson [50] shows that when intrusions are rare, the false-alarm rate becomes the limiting factor for intrusion detection. Our data are the opposite case, since nearly every login attempt is an attack. Gates and Taylor [17] question whether attacks are rare and whether attack-free training data is available, and note that malicious traffic can look normal and normal traffic can look anomalous. Goodall et al. [43] and Best et al. [51] learn a model of each actor's expected behavior, with Best et al. [51] assuming that an actor's activity repeats weekly and aggregating actors for statistical robustness. However, the churning, near-uniformly hostile traffic we observe is unlikely to offer a per-actor normal. Goodall et al. [43] concede that operators "do not want another automated tool with algorithms they do not trust" and keep a human in the loop. Lacking a per-actor baseline, we make campaign identification operator-driven.

D. Username Campaigns

We analyze attackers at the level of a campaign, meaning a group of attackers acting together, rather than one at a time. A campaign often spans multiple IPs. One IP can be assigned to different hosts [52], hosts can cycle through new IPs [53], and counting by IPs can overstate the number of hosts, sometimes by an order of magnitude [54]. An IP-based view is therefore likely to scatter one campaign across many seemingly unrelated entries. However, a campaign's attackers likely draw from a shared credential list, so the same login names may recur across changing IPs. Grouping by username can reassemble the campaign that an IP-by-IP view could split.

E. Operator Labeling and Collaboration

The findings an operator produces can be captured and shared. In two works [24], [55], labeling was used to fold the operator's judgments back into the system. Other systems, such as [22] and [12], supported collaborative analysis through shared artifacts. These systems suggest that labeling and collaboration are valuable, which motivates our labeling-and-sharing mechanism: the step that turns an identified set of attackers into a durable, shared campaign.

III. Tasks and Requirements

The design of SSHVisual draws on our experience as network operators and on gaps in prior work on SSH attack visualization. We derive three tasks and five requirements.

A. Tasks

T1: Identify SSH campaigns. Analysts need to find attackers that behave alike across usernames, IPs, and ASNs, and record groups so they can be referenced.

T2: Detect anomalies that warrant action. Administrators need to spot unusual activity, such as a spike, a dominant source, or low-volume sources that behave like a known attacker, and decide whether to block it.

T3: Share findings across operators. Analysts and administrators need to see what their colleagues have investigated, so that the same patterns are not investigated twice.

B. Requirements

R1: Attacker similarity profiling. SSHVisual should find attackers similar to a username or IP, show shared ASNs, and explain each match. R1 supports T1 and T2: analysts can assemble campaigns, and administrators can find IPs to block.

R2: Sorting and ranking. SSH attack data take many forms, and no single ranking can suit them all. The data can be heavy-tailed: one username can account for a quarter of attacks [56] (in our data, `root` accounts for nearly half), and recurring IPs for over 70% of attempts [57] (in our data, the top 0.34% IP accounts for 0.018% of attacks). The data can also be spread out: distributed attackers split attempts across many low-volume sources, so no entity may stand out by count. SSHVisual should therefore list every entity and let operators sort and rank by any combination of attributes, not only volume. R2 supports T1 and T2: administrators can surface anomalous entities that miss the top of a volume ranking, and analysts can rank by bursts or day-over-day change.

R3: Shared tagging. Operators should be able to tag attackers, record what a tag means and why it was applied, and keep those tags visible to every operator. Each tag application should record who applied it and when, so later operators can judge whether a finding is still current. R3 supports T3 and T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. A table can support precise lookup and ranking but can discard temporal structure; a line chart can preserve it, showing spikes and trends at a glance [58]. SSHVisual should provide both views and let operators move between them. An administrator may find and act on an anomaly using the table alone, whereas an analyst may use the table to assemble related attackers and examine their behavior in the line charts. R4 supports T1 and T2.

R5: Interactive exploration of the line charts. The line charts should follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [59]. They should open on the full time range, let operators zoom and filter entities, and reveal details. R5 supports T1 and T2: zooming can isolate a suspected campaign's window, and filtering can remove dominant entities masking smaller ones. A line chart can become harder to understand as more series are added [60]. Each line chart should therefore show few entities; we cap the number at 10.

IV. Design

SSHVisual combines two visual components (a discovery table and line charts) with two non-visual components (similarity search and tagging). Sections IV-A and IV-B describe

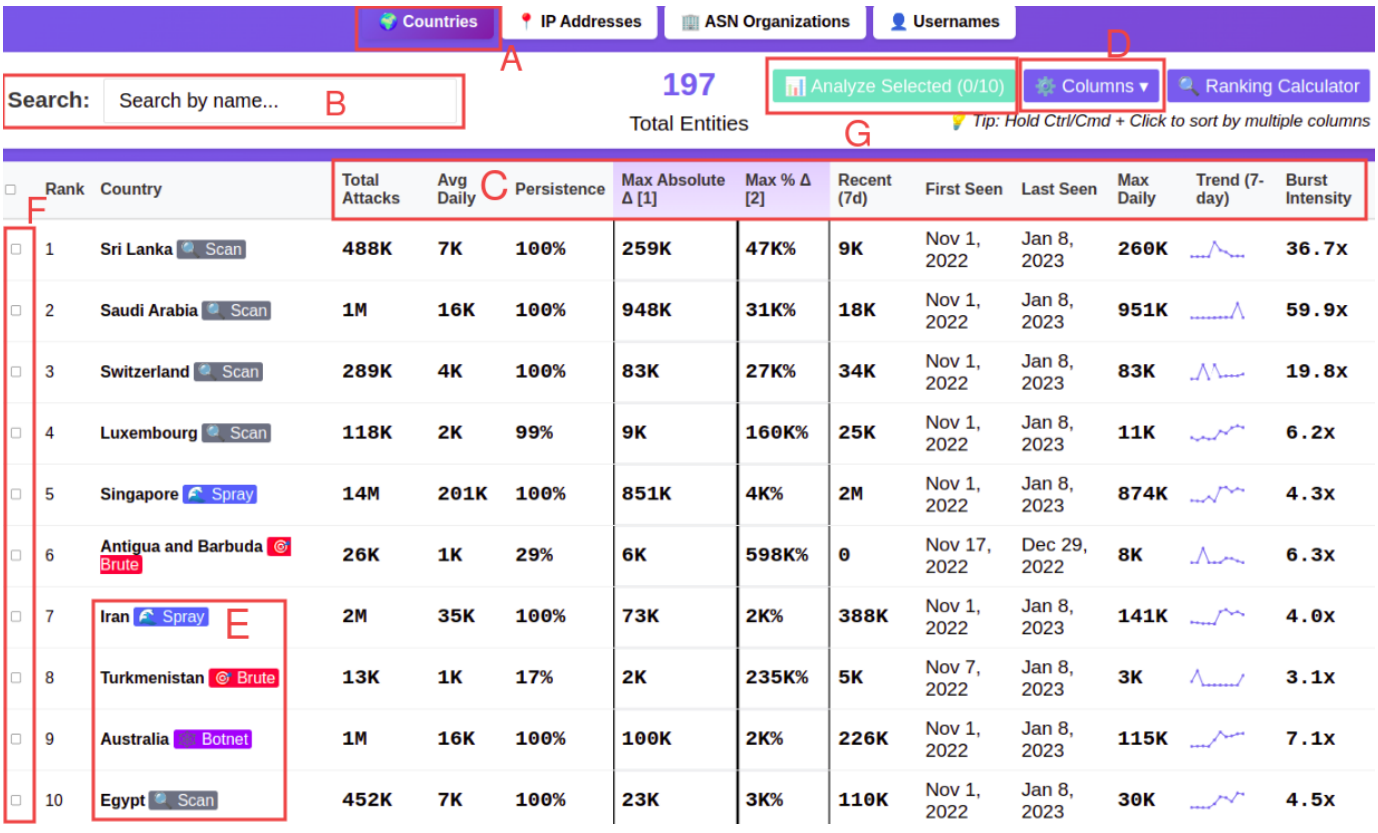


Fig. 1. Discovery table, showing the Countries tab (A). Operators can filter countries (B), sort by any combination of columns (C) and add more columns from the Columns menu (D). A badge (E) summarizes attack patterns. Operators can select up to 10 rows (F) and plot them with Analyze Selected (G).

them, respectively. Our data are 206.6 million failed SSH login attempts on our production systems between Nov. 1, 2022, and Jan. 8, 2023. Each record gives a timestamp, source IP, and attempted username. We map IPs to ASNs and countries.

A. Visual Design

SSHVisual provides two views of the data. The discovery table (Fig. 1) is time-independent: each row summarizes one entity over the whole period. The line charts (Fig. 2) are time-dependent. Both views cover the full data period by default.

1) Discovery Table: The discovery table has four tabs, one per dimension: Countries, IP Addresses, ASN Organizations, and Usernames. In Fig. 1, the Countries tab is active (label A). A search box filters rows (label B), and a counter shows the number of entities (197 countries). Some columns (label C) are shared across all tabs: total attacks, average daily attacks, persistence (the share of days the entity was active), largest absolute and percentage daily change, attacks in the last week, first- and last-seen dates, maximum daily count, trend (a sparkline of attacks at seven-day intervals), and burst intensity (ratio of maximum daily attacks to average daily attacks, where a higher ratio likely indicates sporadic, intense attacks). Appendix A lists all columns (labels C and D). The Countries tab summarizes traffic origins, but because a campaign's hosts can span many countries (Section V), analysis relies mainly on the IP, ASN, and username tabs.

On the Countries tab only, each country has a badge showing its dominant behavior: Brute, Spray, Botnet, or Scan (label E). Brute marks countries which has few IPs making many attacks using few usernames. Spray marks countries with many IPs trying many usernames, and Botnet marks countries with traffic from many IPs and ASNs. Scan marks the rest.

Unlike the line charts, which show the 10 most active entities, the table lists every entity. Operators can select up to 10 entities (label F) and use Analyze Selected (label G) to plot them in the line charts. Operators can thus move from a precise lookup of entities to their temporal behavior.

2) Line Charts: Fig. 2 stacks five line charts on a shared time axis. The first, Total Attacks, shows the aggregate volume (label A); the others show the 10 most active countries, ASNs, IPs, and usernames (labels B–E). Each legend entry (label F) lists the entity's total for the selected period. Interaction with the line charts follows Shneiderman's mantra [59], and the charts are coordinated so that an action in one updates all.

Overview first. The charts cover all 69 days by default, with each chart showing the 10 most active entities in its dimension.

Zoom. Every chart supports brushing (label G marks an example window). When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, keeping all charts in sync. Further brushing is disabled once the range narrows to two days or fewer, since so short a range cannot show a trend.

Filter. Right-clicking a legend entry crosses the entity out, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a

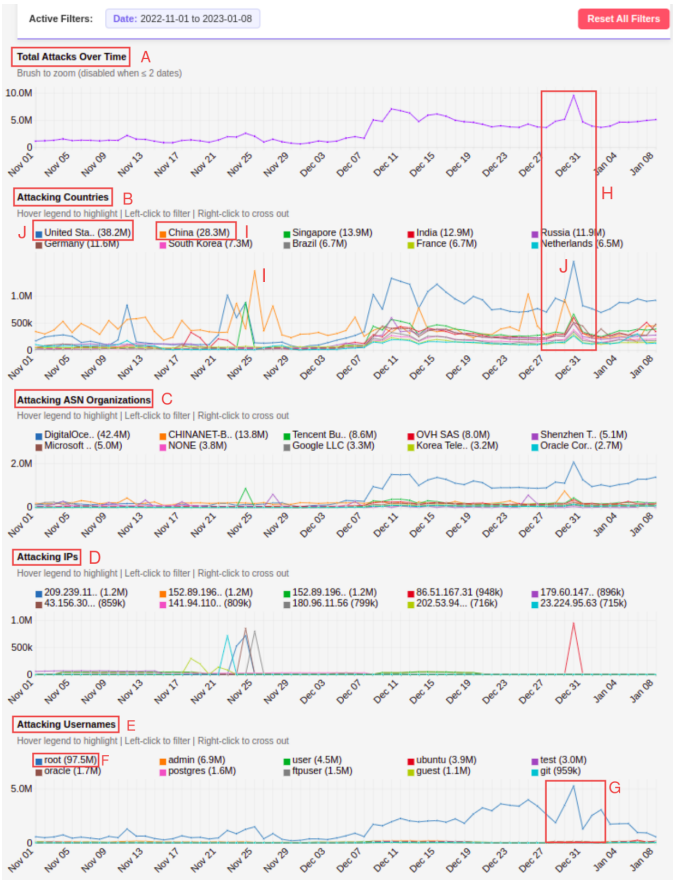


Fig. 2. The top chart (A) shows total attacks over time, and the four charts below it show the 10 most active countries (B), ASNs (C), IPs (D), and usernames (E). A legend entry (F) identifies each series, and (G) marks a window an operator might brush to zoom. A spike appears around Dec. 30 (H), and two lines (I, J) follow individual countries' attacks over time.

dominant one. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack links a username, an IP, an ASN, and a country, the selection propagates along these relationships:

- Country: its ASNs, IPs, and usernames.
- ASN: its countries, IPs, and their usernames.
- IP: its country, ASN, and usernames.
- Username: the countries, ASNs, and IPs that tried it.

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, and hovering over a legend entry highlights the entity's line.

Linked filtering turns four rankings into a means of tracing an entity across dimensions. An operator investigating the Dec. 30 spike (label H) can select `root` in the username chart; the other charts show which IPs, networks, and countries tried `root`, and brushing around Dec. 30 isolates the spike. Conversely, selecting an ASN such as DigitalOcean reveals which usernames its hosts try, which can indicate whether its traffic is one campaign or many. These interactions fulfill R5.

The line charts answer questions that the discovery table cannot, such as "Do the countries in Fig. 1 share the same peaks and valleys?" or "Does one country begin attacking

	Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
☐	1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
☐	2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
☐	3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
☐	4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
☐	5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
☐	6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
☐	7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
☐	8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Fig. 3. Usernames tab ranked by the average of each username's ranks on Max Absolute Δ (A) and Max % Δ (B), surfacing bursty usernames (C).

earlier than another?" In Fig. 2, the aggregate chart shows a spike on Dec. 30 (label H), and most top 10 countries peak on the same day, so the spike is not tied to any one country; the ASN and username charts show DigitalOcean and `root` peaking the same day. The charts also reveal timing differences. China's largest peak falls in late November (label I), weeks before US activity rises in early December and peaks on Dec. 30 (label J), and the IP chart shows IPs that each spike for a day or two, then fall silent. Neither pattern appears in the table's per-entity totals. Together, Figs. 1 and 2 fulfill R4.

3) Sorting and Ranking: Fig. 3 shows the Usernames tab sorted by two columns (labels A and B). Operators add sort columns by holding Ctrl/Cmd while clicking headers. Each entity is ranked separately on every selected column, and the table orders entities by the average of those ranks. The bracketed indices show click order but do not set priority.

Sorting matters because the data can be heavy-tailed. Fig. 2's username chart is dominated by `root`, which accounts for 97.5 million attacks (nearly half of the total), and flattens every other line. Ranking by day-over-day change instead surfaces usernames absent from Fig. 2 (label C in Fig. 3). Several appear to be default accounts for software or hardware products (e.g., `karaf` for Apache Karaf). Their totals are far below the top 10, but their activity is concentrated: `karaf` was used in 78 thousand attacks, 45 thousand of them on a single day, and was active on 15 of the 69 days. Such product-specific bursts suggest campaigns targeting particular devices, yet they can be invisible in a volume-ranked chart. Fig. 3 shows how sorting and ranking over all entities fulfill R2.

B. Grouping and Tagging

The visual components help operators find attackers; grouping and tagging connect them to others and record findings. Similarity search groups usernames by behavior (Section IV-B1) and IPs by behavior or shared usernames (Section IV-B2), and tagging (Section IV-B3) shares these groups.

1) Username Similarity: Similarity search lets an operator start from one username and find usernames that attackers use the same way. The unit of analysis is the population of sources that tries a username. For each username, we compute the breadth of that population (the number of distinct IPs, ASNs, and countries that try it), its volume, and the concentration of its traffic on its most active source, among other features. A diagnostic analysis showed some features which were informative for IPs, turn out uninformative for usernames. Daily stability is an example: most IPs are active on many days, so their stability is well defined, but half of the usernames appear on only one day, for which stability is meaningless. We cluster username profiles with k-means ($k = 9$), selected by silhouette score. Each cluster is labeled automatically from its centroid. To answer a query, SSHVisual retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Fig. 4, an operator searches for `james` (label A). SSHVisual places it in Cluster 3, named High-Volume Persistent Botnet-Scale, with 2,624 usernames (label B). The cluster's typical username is tried in about 31,800 attacks from about 4,000 IPs and is persistent (active on 72% of days) even though the set of IPs trying it changes daily. The results table (label F) lists the 100 most similar usernames (five shown here), each with its total attacks, top source country, top three source IPs, top three ASNs, and a set of Why similar? labels (label D), such as Same lead source, Comparable volume, or Botnet-scale sourcing, that explain the match.

Above the results table, a Shared infrastructure panel (label C) reports 37 IPs that are top sources for more than one returned username. Each panel line names an IP and counts how many of the 100 usernames it is a top source for, followed by examples of those usernames. The two most-shared IPs, `152.89.196.123` and `152.89.196.220`, share a /24 subnet and are top sources for 34 and 33 usernames, respectively, including `manager`, `daniel`, and `oscar`. Among ASNs, 13 are shared, led by DigitalOcean (99 of 100 usernames), Tencent (90), and OVH (40). This panel can turn behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and may belong to the same campaign. Shared IPs are stronger evidence than ASNs, since large providers can host many unrelated attackers.

2) IP Similarity: We characterize each IP by a 16-dimensional profile: 14 features from its activity plus two infrastructure flags. The features fall into four groups: volume (log-scaled totals, daily means, peaks, and final-week attacks); temporal shape (the fraction of days the IP was active, its burst intensity, its largest daily swings, a fitted trend slope, activity span, and the share of traffic in the final week); targeting behavior (the number of unique usernames tried, unique usernames per active day, how stable that set is day to day, and

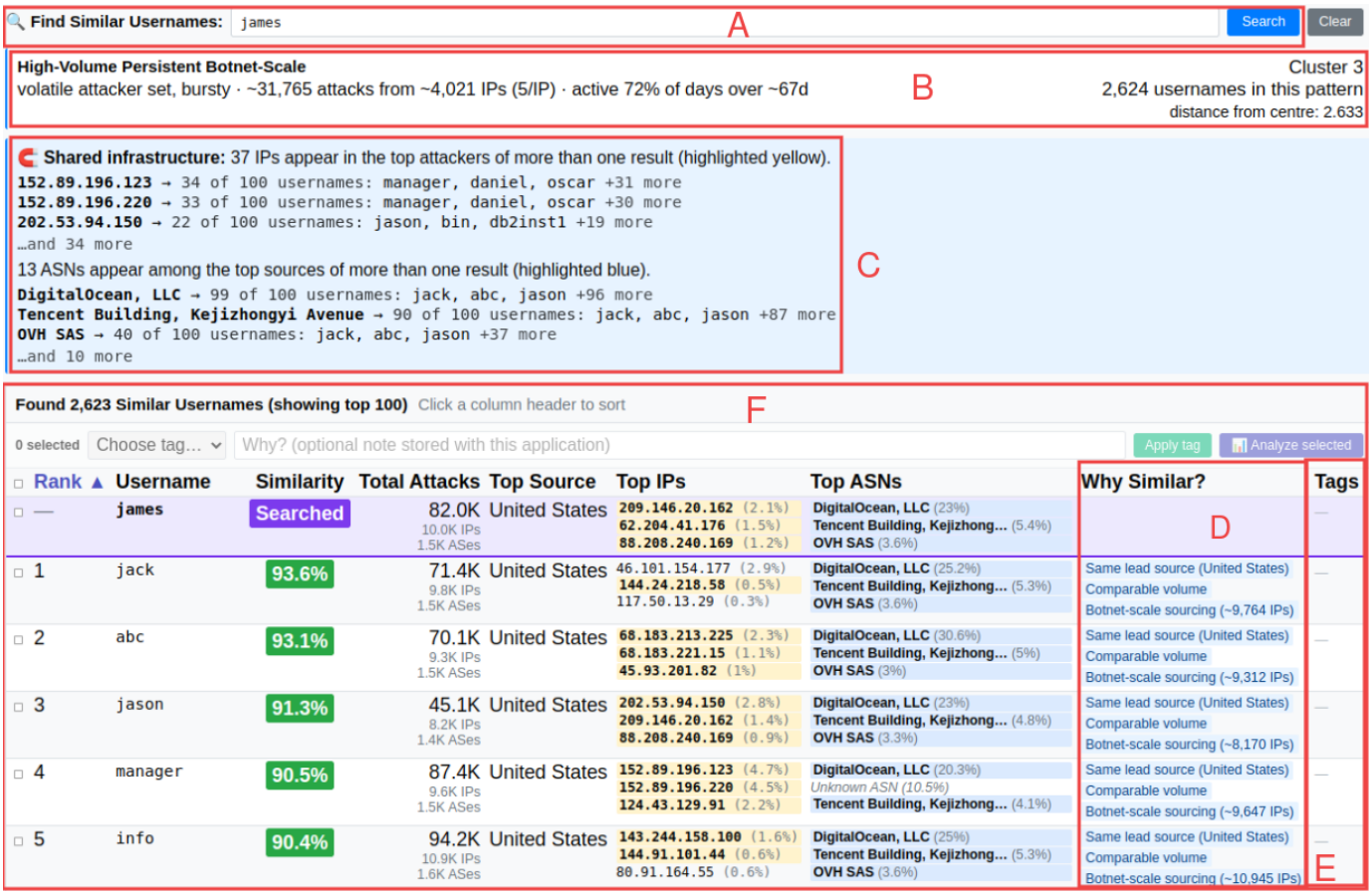


Fig. 4. Similarity search for `james` (A). The cluster summary (B), Shared infrastructure panel (C), and Why similar? labels (D) explain why the results (F) were grouped. The Tags column (E) is described in Section IV-B3.

its top username's share of attacks); and infrastructure (two flags indicating whether the IP belongs to a cloud provider and whether it originates from a major source country). We cluster these profiles with k-means ($k = 20$), selected by silhouette score, label each cluster from its centroid, and answer queries as for usernames.

IP similarity search offers two modes, selected with the Find by toggle (label C in Fig. 5). The *behavioral* mode, described above, ranks the other members of the query's cluster by their distance to the query. The *shared-username* mode instead compares the query against every IP in the data set, and scores each IP by the overlap between its usernames and the query's. Overlap on uncommon usernames (tried by fewer than 1% of IPs) is weighted more heavily than overlap on common ones such as `root`, because unrelated attackers are unlikely to share them by chance. For each result, the table reports the number of shared usernames and shared uncommon usernames. The behavioral mode finds IPs that attack in the same way, whereas the shared-username mode finds IPs that work from the same credential list, even when their volume or timing differs.

In Fig. 5, an operator searches for `61.177.173.46` (label A), which falls in Cluster 5 (Low-Volume Persistent Exploratory, 4,405 IPs). The most similar results (label B) score between 80.4% and 91.2%, and all belong to CHINANET-BACKBONE. The cluster's name describes its centroid; the query and its neighbors are atypical members, each sending over 480 thousand attempts with a single username (`root`). Although the profile contains no IP or ASN features, it grouped these hosts, suggesting a common operator.

Figs. 4 and 5 show how and why similar attackers are grouped along the username and IP dimensions, fulfilling R1.

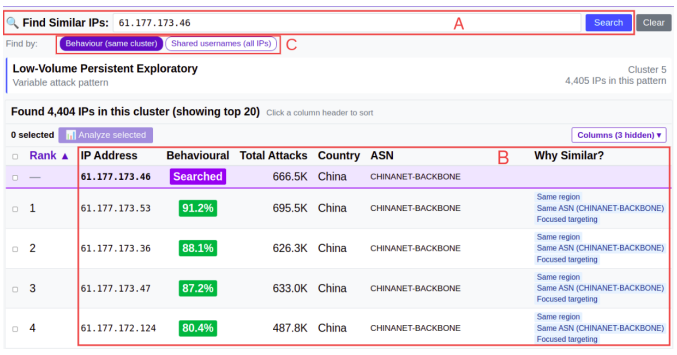


Fig. 5. Similarity search for IP `61.177.173.46` (A) in behavioral mode, one of two modes (C). Neighbors are nearby IPs in the same ASN (B).

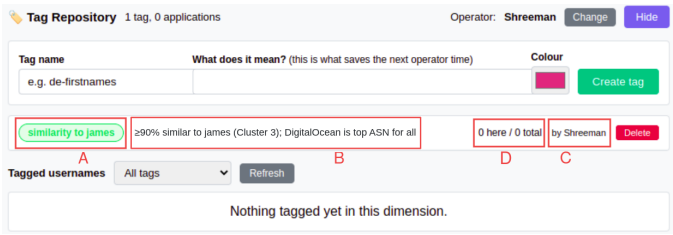


Fig. 6. A newly created tag (A), listed with its description (B), author (C), and application counts in this dimension (D).

3) Shared Tagging: Figs. 6–8 show how operators record findings for others. The tag repository records the author of every tag and application.

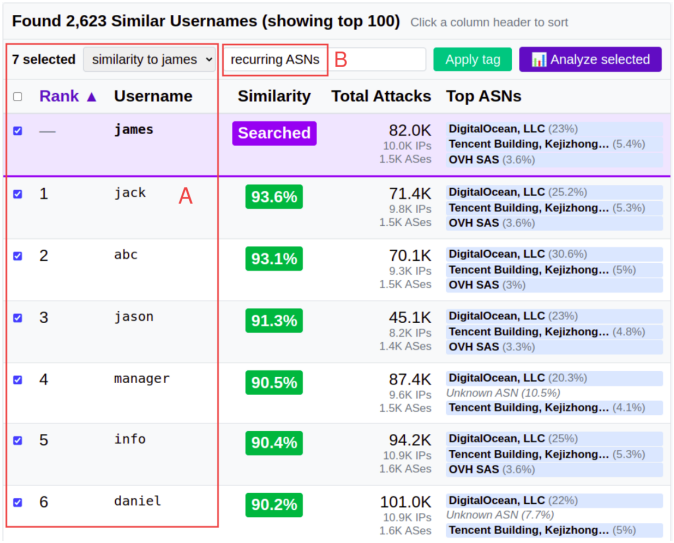


Fig. 7. Applying the tag to seven usernames from the similarity results (A) of Fig. 4, with a note explaining the application (B).

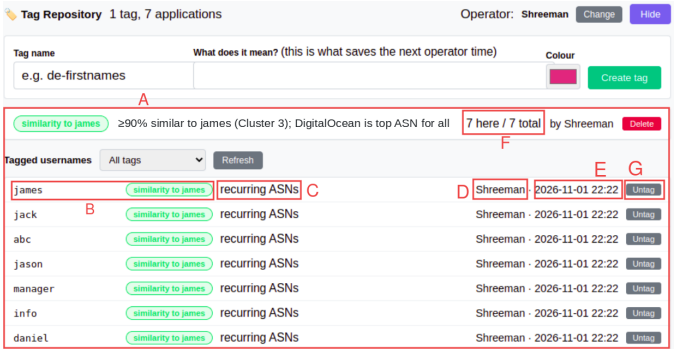


Fig. 8. Repository after tagging (A), listing the usernames (B) with note (C), author (D), timestamp (E), application counts (F), and Untag (G).

In Fig. 6, the operator creates a tag, similarity to james (label A), for the usernames found in Fig. 4. The repository lists the tag with its description (label B) and author (label C). In Fig. 7, the operator selects seven usernames (label A), chooses the new tag, and adds the note recurring ASNs (label B). In Fig. 8, the repository reports seven applications (labels A and F). Below, the list shows each tagged username with its note (labels B and C), author (label D), and a timestamp (label E); any application can be removed with Untag (label G). Tagged usernames also appear in the Tags column of future similarity searches (label E in Fig. 4; empty there because the screenshot predates tagging), so an operator who later retrieves them sees they were already investigated.

Separating a tag's meaning from each application's rationale is intended to let later operators understand both what a group represents and why each member belongs, without repeating the analysis. The timestamp indicates whether a finding is recent, and the author shows whom to ask. Figs. 6–8 thus show how tagging lets others build on findings, fulfilling R3.

V. Evaluation

We demonstrate how SSHVisual uncovers hidden patterns and entities through two case studies conducted by the authors: 1) analysts finding a campaign where the same usernames recur across IPs in different countries (Section V-A), and 2) administrators finding similar entities to block that are buried beneath higher-volume ones (Section V-B). Each study starts with an operator finding an IP, e.g., from an intrusion-detection alert, abuse report, or the discovery table. Alone, an IP is one among many; SSHVisual can help find its connections.

A. Security Analysts: Username Campaigns

Fig. 9 shows the shared-username search (label A) for IP `68.183.213.225` (DigitalOcean, Germany). Five IPs match at 100% (labels B–F): `143.244.183.240`, `24.199.108.32`, and `146.190.113.228` in the US, `134.122.62.125` in the Netherlands, and `101.204.27.20` in China. All six try the same 77 usernames, including 16 uncommon ones such as `ecell`, `guojing`, and `xxaifwq`. A 100% score built on common usernames such as `root` and `admin` would say little, but 16 uncommon usernames are unlikely to be shared by chance. The six IPs share two ASNs but span four countries, consistent with a single operator controlling hosts in several regions. In total, 12 IPs (including these five) score above 84%, spanning five countries and six ASNs (Appendix B).

Fig. 10 shows why volume-based views may miss this campaign. Each IP peaks on days the others do not: `101.204.27.20` on Nov. 25 (label F), `146.190.113.228` on Dec. 19 (label D), `143.244.183.240` on Dec. 20 (label B), `24.199.108.32` on Dec. 28 (label C), `68.183.213.225` on Jan. 2 and 8 (label A), and `134.122.62.125` on Jan. 5–6 (label E). The six labeled lines' main peaks never coincide, so the chart suggests unrelated bursts. Shared usernames suggest one campaign.

B. Network Administrators: Finding Hidden Entities to Block

Fig. 11 shows the behavioral IP similarity search for `200.195.169.59`, an IP from Ligga Telecomunicações S.A. in Brazil with 36 thousand attacks. Its five closest matches

Rank	IP Address	Usernames	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	68.183.213.225 A	—	its top usernames ecell guojing xxaifwq +74 more	16 its uncommon usernames ecell guojing xxaifwq +13 more	373.6K	Germany	DigitalOcean, LLC
1	143.244.183.240 B	100%	ecell guojing xxaifwq +74 more	16 ecell guojing xxaifwq +13 more	136.4K	United States	DigitalOcean, LLC
2	24.199.108.32 C	100%	guojing escheduler ecell +74 more	16 guojing escheduler ecell +13 more	140.5K	United States	DigitalOcean, LLC
3	146.190.113.228 D	100%	guojing ecell xxaifwq +74 more	16 guojing ecell xxaifwq +13 more	155.4K	United States	DigitalOcean, LLC
4	134.122.62.125 E	100%	ecell escheduler guojing +74 more	16 ecell escheduler guojing +13 more	159.1K	Netherlands	DigitalOcean, LLC
5	101.204.27.20 F	100%	ecell guojing xxaifwq +74 more	16 ecell guojing xxaifwq +13 more	75.5K	China	CHINA UNICOM China169 Backbone

Fig. 9. The searched IP (A) and five IPs with 100% shared-username similarity (B–F). Labels A–F mark the same IPs in Fig. 10.

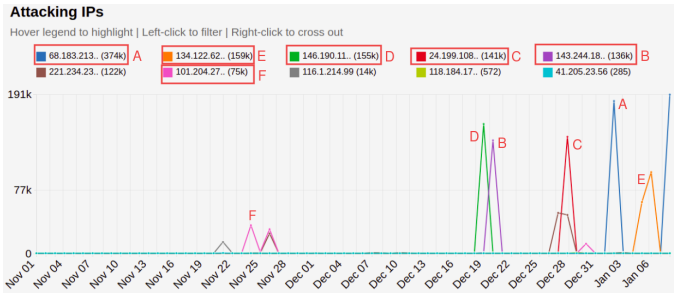


Fig. 10. Line chart of the six IPs (A–F) from Fig. 9, along with other IPs that have nearly identical username lists.

(88.6–94.9% similar) come from Brazil, Indonesia, Italy, Mexico, and Australia, each from a different ASN. Despite this spread, they behave alike. Each made roughly 30-40 thousand attacks and is labeled as Exploratory targeting, meaning it tries many different usernames. Although the behavioral mode does not compare usernames, each match shares 1,684 to 1,899 usernames with the searched IP, 46 to 140 of them uncommon (tried by fewer than 1% of IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. The pattern extends beyond these five: the top 20 matches, all at least 86% similar, span 14 countries and 20 ASNs (Appendix C). Together, they suggest relatively low-volume IPs in many countries are working through large, overlapping username lists.

Fig. 12 shows how deeply these IPs are buried, though administrators may want to block them. None of the 10 IPs (the searched IP and its nine closest matches; Fig. 11 shows the first five and Appendix C the rest) would appear in the top 10 IP chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6 million attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN would need 13 times its volume to reach the top 10 ASNs, and the IP would need 20 times its volume to reach the top 10 IPs. The other IPs rank between #222 and #2,826. An administrator looking only at the top 10 charts would see

Rank	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	200.195.169.59	Searched	its top usernames oix jqz alu +3818 more	925 its uncommon usernames oix jqz alu +922 more	36.4K	Brazil	Ligga Telecomunicações S.A.
1	132.255.253.123	94.9%	nw noroot elloise +1799 more	46 elloise alameda tab +43 more	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
2	103.186.99.250	92%	final maison virginie +1681 more	72 final maison virginie +69 more	29.5K	Indonesia	PT Network Center Nusantara
3	79.3.19.34	90.8%	sdt mallika yfc +1896 more	83 sdt helix cch +80 more	36.2K	Italy	Telecom Italia S.p.A.
4	201.116.12.217	89.2%	nash mule vojta +1863 more	140 nash vojta teamspeak-server +137 more	33.8K	Mexico	UNINET
5	27.99.9.141	88.6%	anusha andri prabhu +1839 more	86 damares milo libevent +83 more	41.2K	Australia	Microplex PTY LTD

Fig. 11. The five closest behavioral matches to `200.195.169.59`.

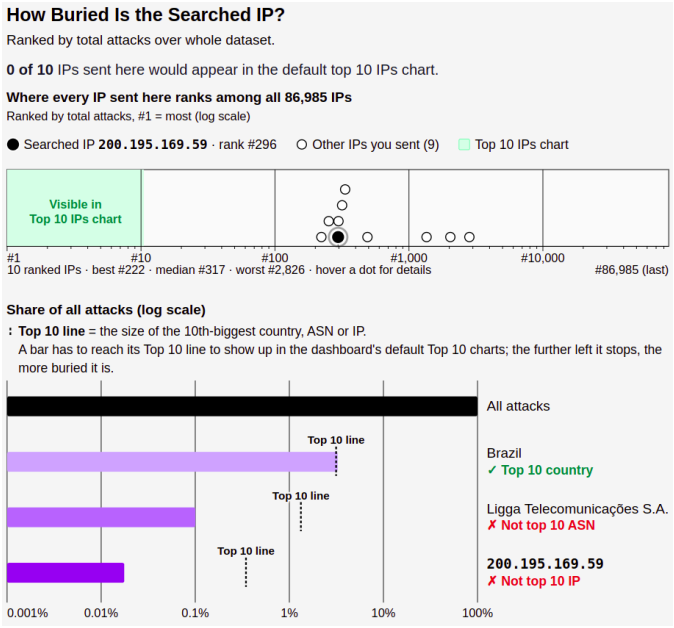


Fig. 12. Top 10 chart does not include `200.195.169.59` or its matches.

Brazil but never these IPs. With SSHVisual, administrators can learn not only which IP to block but also which similar IPs they may want to block and, most importantly, why.

VI. Conclusion

We presented SSHVisual, a visual analytics system that treats the username as a first-class dimension alongside the IP and ASN in SSH attack data. SSHVisual pairs a discovery table that ranks every entity with coordinated line charts that follow Shneiderman's mantra, so operators can move between precise lookup and temporal behavior. Similarity search groups attackers by behavior or by the usernames they share, and explains each match. Shared tagging preserves those groups so findings persist and remain visible to all operators. Our case studies suggest this combination can surface patterns

that volume-based, IP-centric views can miss. Relatively low-volume IPs, spread across many countries and ASNs, worked through large, overlapping username lists, yet none ranked high enough to appear in any top 10 chart. IPs peaking on different days were linked by the 77 usernames they shared into what appears to be a single campaign. Because SSH traffic may not offer a normal baseline for automated detection, these results suggest that exploratory, operator-driven analysis remains a useful complement to automated methods, and that usernames can be a valuable lens for recognizing coordinated attackers. Our evaluation consists of case studies conducted by the authors on 69 days of data from our own systems; a study with external operators, and evaluation on other data sets and time periods, are needed to confirm these findings.

REFERENCES

[1] T. Ylonen, "Ssh–secure login connections over the internet," in *Proceedings of the 6th USENIX Security Symposium*, vol. 37, 1996, pp. 40–52.

[2] S. K. Singh, S. Gautam, C. Cartier, S. Patil, and R. Ricci, "Where the wild things are: Brute-Force SSH attacks in the wild and how to stop them," in *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*. Santa Clara, CA: USENIX Association, Apr. 2024, pp. 1731–1750. [Online]. Available: https://www.usenix.org/conference/nsdi24/presentation/singh-sachin

[3] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis *et al.*, "Understanding the mirai botnet," in *26th USENIX security symposium (USENIX Security 17)*, 2017, pp. 1093–1110.

[4] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "Ddos in the iot: Mirai and other botnets," *computer*, vol. 50, no. 7, pp. 80–84, 2017.

[5] F. Dang, Z. Li, Y. Liu, E. Zhai, Q. A. Chen, T. Xu, Y. Chen, and J. Yang, "Understanding fileless attacks on linux-based iot devices with honeycloud," in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019, pp. 482–493.

[6] S. Pastrana and G. Suarez-Tangil, "A first look at the crypto-mining malware ecosystem: A decade of unrestricted wealth," in *Proceedings of the Internet Measurement Conference*, 2019, pp. 73–86.

[7] V. T. Guimarães, C. M. D. S. Freitas, R. Sadre, L. M. R. Tarouco, and L. Z. Granville, "A survey on information visualization for network and service management," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 285–323, 2016.

[8] H. Shiravi, A. Shiravi, and A. A. Ghorbani, "A survey of visualization systems for network security," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 8, pp. 1313–1329, 2012.

[9] T. Zhang, X. Wang, Z. Li, F. Guo, Y. Ma, and W. Chen, "A survey of network anomaly visualization," *Science China Information Sciences*, vol. 60, no. 12, p. 121101, 2017.

[10] J. Raynor, T. Crnovrsanin, S. Di Bartolomeo, L. South, D. Saffo, and C. Dunne, "The state of the art in bgp visualization tools: A mapping of visualization techniques to cyberattack types," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 1059–1069, 2022.

[11] C. Wu, S. Sheng, and X. Dong, "Research on visualization systems for ddos attack detection," in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2018, pp. 2986–2991.

[12] J. R. Goodall and M. Sowul, "Viassist: Visual analytics for cyber defense," in *2009 IEEE conference on technologies for homeland security*. IEEE, 2009, pp. 143–150.

[13] C. Estan and G. Varghese, "New directions in traffic measurement and accounting," in *Proceedings of the ACM SIGCOMM 2002 Conference*. ACM, 2002, pp. 323–336.

[14] S. Staniford, J. A. Hoagland, and J. M. McAlerney, "Practical automated detection of stealthy portscans," *Journal of Computer Security*, vol. 10, no. 1–2, pp. 105–136, 2002.

[15] H. Griffioen and C. Doerr, "Discovering collaboration: Unveiling slow, distributed scanners based on common header field patterns," in *NOMS 2020 – IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2020, pp. 1–9.

[16] G. Gu, R. Perdisci, J. Zhang, and W. Lee, "BotMiner: Clustering analysis of network traffic for protocol- and structure-independent botnet detection," in *Proceedings of the 17th USENIX Security Symposium*. USENIX Association, 2008, pp. 139–154.

[17] C. Gates and C. Taylor, "Challenging the anomaly detection paradigm: A provocative discussion," in *Proceedings of the 2006 workshop on New security paradigms*, 2006, pp. 21–29.

[18] F. Mansmann and S. Mansmann, "Interactive exploration of data traffic with hierarchical network maps," 2006.

[19] J. Oberheide, M. Goff, and M. Karir, "Flamingo: Visualizing internet traffic," in *2006 IEEE/IFIP Network Operations and Management Symposium NOMS 2006*. IEEE, 2006, pp. 150–161.

[20] F. Mansmann, F. Fischer, D. A. Keim, and S. C. North, "Visual support for analyzing network traffic and intrusion detection events using treemap and graph representations," in *Proceedings of the Symposium on Computer Human Interaction for the Management of Information Technology*, 2009, pp. 19–28.

[21] T. Taylor, D. Paterson, J. Glanfield, C. Gates, S. Brooks, and J. McHugh, "Flovis: Flow visualization system," in *2009 Cybersecurity Applications & Technology Conference for Homeland Security*. IEEE, 2009, pp. 186–198.

[22] S. Chen, C. Guo, X. Yuan, F. Merkle, H. Schaefer, and T. Ertl, "Oceans: Online collaborative explorative analysis on network security," in *Proceedings of the Eleventh Workshop on Visualization for Cyber Security*, 2014, pp. 1–8.

[23] T. Tremel, J. Kogel, F. Jauernig, S. Meier, D. Thom, F. Becker, C. Muller, and S. Koch, "Vitalflow: Visual interactive traffic analysis with netflow," in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2022, pp. 1–6.

[24] L. Xiao, J. Gerth, and P. Hanrahan, "Enhancing visual analysis of network traffic using a knowledge representation," in *2006 IEEE symposium on visual analytics science and technology*. IEEE, 2006, pp. 107–114.

[25] D. H. Jeong, J.-H. Cho, F. Chen, L. Kaplan, A. Jøsang, and S.-Y. Ji, "Interactive web-based visual analysis on network traffic data," *Information*, vol. 14, no. 1, p. 16, 2022.

[26] H. Choi, H. Lee, and H. Kim, "Fast detection and visualization of network attacks on parallel coordinates," *computers & security*, vol. 28, no. 5, pp. 276–288, 2009.

[27] Y. Livnat, J. Agutter, S. Moon, and S. Foresti, "Visual correlation for situational awareness," in *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.* IEEE, 2005, pp. 95–102.

[28] C. Humphries, N. Prigent, C. Bidan, and F. Majorczyk, "Elvis: Extensible log visualization," in *Proceedings of the Tenth Workshop on Visualization for Cyber Security*, 2013, pp. 9–16.

[29] T. Yu, R. Lippmann, J. Riordan, and S. Boyer, "Ember: a global perspective on extreme malicious behavior," in *Proceedings of the seventh international symposium on visualization for cyber security*, 2010, pp. 1–12.

[30] D. H. Jeong, B.-K. Jeong, and S.-Y. Ji, "Multi-resolution analysis with visualization to determine network attack patterns," *Applied Sciences*, vol. 13, no. 6, p. 3792, 2023.

[31] A. King, B. Huffaker, A. Dainotti, and K. Claffy, "A coordinated view of the temporal evolution of large-scale internet events," *Computing*, vol. 96, no. 1, pp. 53–65, 2014.

[32] D. Ceneda, M. Di Bartolomeo, V. Di Donato, M. Patrignani, M. Pizzonia, and M. Rimondini, "Routingwatch: Visual exploration and analysis of routing events," in *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2016, pp. 591–597.

[33] J. C. Roberts, "State of the art: Coordinated & multiple views in exploratory visualization," in *Fifth international conference on coordinated and multiple views in exploratory visualization (CMV 2007)*. IEEE, 2007, pp. 61–71.

[34] J. S. Yi, Y.-a. Kang, J. T. Stasko, and J. A. Jacko, "Understanding and characterizing insights: how do people gain insights using information visualization?" in *Proceedings of the 2008 Workshop on BEyond time and errors: novel evaLuation methods for Information Visualization*, 2008, pp. 1–6.

[35] R. Djap, C. Lim, K. E. Silaen, and A. Yusuf, "Xb-pot: Revealing honeypot-based attacker's behaviors," in *2021 9th International Conference on Information and Communication Technology (ICoICT)*. IEEE, 2021, pp. 550–555.

[36] V. Visoottiviseth, U. Jaralrungroj, E. Phoomrungraungsuk, and P. Kultanon, "Distributed honeypot log management and visualization of attacker geographical distribution," in *2011 Eighth International Joint*

Conference on Computer Science and Software Engineering (JCSSE). IEEE, 2011, pp. 23–28.

[37] M. Baykara and R. Das, "A novel honeypot based security approach for real-time intrusion detection and prevention systems," *Journal of Information Security and Applications*, vol. 41, pp. 103–116, 2018.

[38] A. Belqruch and A. Maach, "Scada security using ssh honeypot," in *Proceedings of the 2nd International Conference on Networking, Information Systems & Security*, 2019, pp. 1–5.

[39] I. Koniaris, G. Papadimitriou, and P. Nicopolitidis, "Analysis and visualization of ssh attacks using honeypots," in *Eurocon 2013*. IEEE, 2013, pp. 65–72.

[40] A. Kyriakou and N. Sklavos, "Container-based honeypot deployment for the analysis of malicious activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*. IEEE, 2018, pp. 1–4.

[41] I. Koniaris, G. Papadimitriou, P. Nicopolitidis, and M. Obaidat, "Honeypots deployment for the analysis and visualization of malware activity and malicious connections," in *2014 IEEE international conference on communications (ICC)*. IEEE, 2014, pp. 1819–1824.

[42] D. Fraunholz, M. Zimmermann, S. D. Anton, J. Schneider, and H. D. Schotten, "Distributed and highly-scalable wan network attack sensing and sophisticated analysing framework based on honeypot technology," in *2017 7th International Conference on Cloud Computing, Data Science & Engineering-Confluence*. IEEE, 2017, pp. 416–421.

[43] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 204–214, 2018.

[44] Q. Liao and A. Striegel, "Intelligent network management using graph differential anomaly visualization," in *2012 IEEE Network Operations and Management Symposium*. IEEE, 2012, pp. 1008–1014.

[45] W. A. Pike, C. Scherrer, and S. Zabriskie, "Putting security in context: Visual correlation of network activity with real-world information," in *VizSEC 2007: Proceedings of the Workshop on Visualization for Computer Security*. Springer, 2008, pp. 203–220.

[46] L. Hao, C. G. Healey, and S. E. Hutchinson, "Ensemble visualization for cyber situation awareness of network security data," in *2015 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2015, pp. 1–8.

[47] F. Pouget, G. Urvoy-Keller, and M. Dacier, "Time signatures to detect multi-headed stealthy attack tools," in *18th Annual FIRST Conference*, Baltimore, MD, USA, 2006.

[48] O. Thonnard and M. Dacier, "A framework for attack patterns' discovery in honeynet data," *Digital Investigation*, vol. 5, pp. S128–S139, 2008, proc. DFRWS 2008.

[49] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *2010 IEEE symposium on security and privacy*. IEEE, 2010, pp. 305–316.

[50] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security (TISSEC)*, vol. 3, no. 3, pp. 186–205, 2000.

[51] D. M. Best, R. P. Hafen, B. K. Olsen, and W. A. Pike, "Atypical behavior identification in large-scale network traffic," in *2011 IEEE Symposium on Large Data Analysis and Visualization*. IEEE, 2011, pp. 15–22.

[52] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are ip addresses?" in *Proceedings of the 2007 conference on Applications, technologies, architectures, and protocols for computer communications*, 2007, pp. 301–312.

[53] B. Stone-Gross, M. Cova, L. Cavallaro, B. Gilbert, M. Szydlowski, R. Kemmerer, C. Kruegel, and G. Vigna, "Your botnet is my botnet: analysis of a botnet takeover," in *Proceedings of the 16th ACM conference on Computer and communications security*, 2009, pp. 635–647.

[54] G. C. Moura, C. Ganán, Q. Lone, P. Poursaied, H. Asghari, and M. van Eeten, "How dynamic is the isps address space? towards internet-wide dhcp churn estimation," in *2015 IFIP Networking Conference (IFIP Networking)*. IEEE, 2015, pp. 1–9.

[55] J. L. Guerra, E. Veas, and C. A. Catania, "A study on labeling network hostile behavior with intelligent interactive tools," in *2019 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2019, pp. 1–10.

[56] J. Owens and J. Matthews, "A study of passwords and methods used in brute-force SSH attacks," in *USENIX Workshop on Large-Scale Exploits and Emergent Threats (LEET)*, San Francisco, CA, USA, 2008. [Online]. Available: https://lin-web.clarkson.edu/~jmatthew/publications/leet08.pdf

[57] Y. Wu, P. M. Cao, A. Withers, Z. T. Kalbarczyk, and R. K. Iyer, "Mining threat intelligence from billion-scale SSH brute-force attacks," in *Workshop on Decentralized IoT Systems and Security (DISS)*. San Diego, CA, USA: Internet Society, 2020. [Online]. Available: https://www.ndss-symposium.org/wp-content/uploads/2020/04/diss2020-23007-paper.pdf

[58] S. Few, "Visual communication: Core design principles for displaying quantitative information," Perceptual Edge whitepaper, 2006. [Online]. Available: https://public.dhe.ibm.com/software/data/sw-library/cognos/pdfs/whitepapers/wp_visual_communications_core_design_principles.pdf

[59] B. Shneiderman, "The eyes have it: A task by data type taxonomy for information visualizations," in *Proceedings of the IEEE Symposium on Visual Languages*, 1996, pp. 336–343.

[60] W. Javed, B. McDonnel, and N. Elmqvist, "Graphical perception of multiple time series," *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 927–934, 2010.

APPENDIX A
DISCOVERY TABLE COLUMNS

This appendix lists every column of the discovery table described in Section IV-A1. That section shows the Countries tab (Fig. 1) and describes only the columns shared by all four tabs (label C in Fig. 1). The remaining columns are added from the Columns menu (label D in Fig. 1) and depend on the tab. Table I lists every column and the tabs in which it appears, and Figs. 13–16 show all columns of the Countries, IP Addresses, ASN Organizations, and Usernames tabs, respectively.

Columns shared by all tabs. The upper part of Table I lists the ten columns described in Section IV-A1: total attacks, average daily attacks, persistence, largest absolute and percentage daily change, attacks in the last week, first- and last-seen dates, maximum daily count, trend, and burst intensity.

Cross-dimension columns. The lower part of Table I lists columns that describe an entity in terms of another dimension. For each of the four dimensions (usernames, IPs, ASNs, and countries), a tab can have three such columns:

- *Unique count*: the number of distinct values of that dimension associated with the entity, e.g., how many distinct usernames a country's IPs tried.
- *Stability*: the mean Jaccard similarity between the sets of values of that dimension on consecutive days, from 0 (volatile) to 1 (stable). A higher value means the entity uses the same values from day to day.
- *Concentration*: the top three values of that dimension and their percentages of the entity's attacks.

A tab never has these columns for its own dimension; for example, the Usernames tab has no username stability column. The IP Addresses tab also has no ASN or country columns of this kind, because each IP is assumed to belong to exactly one ASN and one country.

Rank	Country	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique ASNs	Unique IPs	Unique Usernames	ASN Stability	IP Stability	Username Stability	ASN Concentration (Top 3)	IP Concentration (Top 3)	Username Concentration (Top 3)	Trend (7-day)	Burst Intensity
1	United States Spray	38.2M	553.0K	100.00% (69d)	748.4K	342.86%	6.0M	Nov 1, 2022	Jan 8, 2023	1.6M	661	13.2K	82.0K	0.705	0.711	0.479	DigitalOcean, LLC (42.9%) Tencent Building, Kejizhongyi Avenue (7.3%) Microsoft Corporation (5.3%)	209.239.112.213 (3.3%) 23.224.95.63 (1.9%) 98.159.98.150 (1.6%)	root (45.6%) admin (3.1%) ubuntu (1.7%)		3.0x
2	China Spray	28.3M	410.1K	100.00% (69d)	1.1M	270.43%	2.0M	Nov 1, 2022	Jan 8, 2023	1.5M	117	10.2K	34.7K	0.814	0.498	0.418	CHINANET-BACKBONE (48.8%) Shenzhen Tencent Computer Systems Company Limited (18.1%) CHINA UNICOM China169 Backbone (7.9%)	180.96.11.56 (2.8%) 61.177.173.53 (2.5%) 61.177.173.49 (2.4%)	root (73.6%) admin (2.9%) user (2.2%)		3.6x
3	Singapore Spray	13.9M	201.2K	100.00% (69d)	850.7K	3686.05%	2.4M	Nov 1, 2022	Jan 8, 2023	873.7K	72	3.7K	51.4K	0.848	0.763	0.466	DigitalOcean, LLC (59.2%) Tencent Building, Kejizhongyi Avenue (22.3%) GoDaddy.com, LLC (3.5%)	43.156.30.199 (6.2%) 148.72.247.138 (1.4%) 157.230.244.66 (0.5%)	root (48.9%) admin (2.1%) ubuntu (1.5%)		4.3x

Fig. 13. All columns of the Countries tab.

Rank	IP Address	Total Attacks	Avg Daily [1]	Persistence [2]	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique Usernames	Username Concentration (Top 3)	Username Stability	Trend (7-day)	Burst Intensity
1	61.177.173.53	695.5K	10.1K	100.00% (69d)	8.1K	152.20%	36.9K	Nov 1, 2022	Jan 8, 2023	18.2K	1	root (100.0%)	1.000		1.8x
2	61.177.173.49	674.0K	9.8K	100.00% (69d)	9.1K	15.44K%	40.4K	Nov 1, 2022	Jan 8, 2023	20.7K	1	root (100.0%)	1.000		2.1x
3	61.177.173.47	633.0K	9.2K	100.00% (69d)	7.4K	111.83%	36.7K	Nov 1, 2022	Jan 8, 2023	16.6K	1	root (100.0%)	1.000		1.8x
4	61.177.173.50	611.8K	8.9K	100.00% (69d)	10.5K	26.95K%	34.8K	Nov 1, 2022	Jan 8, 2023	20.4K	1	root (100.0%)	1.000		2.3x

Fig. 14. All columns of the IP Addresses tab.

Rank	ASN	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Country Concentration (Top 3)	Country Stability	Unique IPs	IP Concentration (Top 3)	Unique Usernames	Username Concentration (Top 3)	IP Stability	Username Stability	Trend (7-day)	Burst Intensity
1	DigitalOcean, LLC	42.4M	80.8K	100.00% (69d)	810.8K	251.22K%	8.1M	Nov 1, 2022	Jan 8, 2023	829.6K	United States (38.6%) Singapore (19.4%) India (13.0%)	0.987	6.0K	68.183.213.225 (0.9%) 143.244.158.100 (0.6%) 68.183.221.15 (0.4%)	61.4K	root (45.3%) admin (1.9%) ubuntu (1.9%)	0.862	0.537		10.3x
2	CHINANET-BACKBONE	13.8M	200.2K	100.00% (69d)	553.1K	329.67%	775.5K	Nov 1, 2022	Jan 8, 2023	751.5K	China (100.0%)	1.000	3.5K	61.177.173.53 (5.0%) 61.177.173.49 (4.9%) 61.177.173.46 (4.8%)	1.7K	root (91.2%) admin (1.2%) user (0.8%)	0.416	0.578		3.8x
3	Tencent Building, Kejizhongyi Avenue	8.6M	13.2K	100.00% (69d)	853.5K	389.72K%	1.5M	Nov 1, 2022	Jan 8, 2023	853.7K	Singapore (35.9%) United States (32.4%) Germany (14.0%)	0.967	1.7K	43.156.30.199 (10.0%) 49.51.19.119 (0.5%) 43.131.41.251 (0.4%)	47.8K	root (51.3%) admin (1.5%) ubuntu (1.3%)	0.805	0.422		64.6x

Fig. 15. All columns of the ASN Organizations tab.

Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique Countries	Unique ASNs	Unique IPs	Country Concentration (Top 3)	ASN Concentration (Top 3)	IP Concentration (Top 3)	Country Stability	ASN Stability	IP Stability	Trend (7-day)	Burst Intensity
1	root	97.5M	1.4M	100.00% (69d)	1.8M	194.38%	10.8M	Nov 1, 2022	Jan 8, 2023	5.2M	192	4.5K	67.5K	China (21.4%) United States (17.8%) Singapore (7.0%)	DigitalOcean, LLC (19.7%) CHINANET-BACKBONE (12.9%) Tencent Building, Kejizhongyi Avenue (4.5%)	86.51.167.31 (1.0%) 43.156.30.199 (0.9%) 180.96.11.56 (0.8%)	0.906	0.692	0.590		3.7x
2	admin	6.9M	99.4K	100.00% (69d)	83.6K	140.68%	919.0K	Nov 1, 2022	Jan 8, 2023	195.5K	191	4.1K	47.7K	United States (17.3%) Russia (13.2%) China (12.0%)	DigitalOcean, LLC (12.0%) NONE (6.4%) Google LLC (4.9%)	91.212.166.22 (3.0%) 152.89.196.220 (2.6%) 152.89.196.123 (2.6%)	0.831	0.541	0.371		2.0x
3	user	4.5M	64.6K	100.00% (69d)	61.2K	151.86%	499.5K	Nov 1, 2022	Jan 8, 2023	142.1K	188	3.9K	41.7K	Lithuania (15.2%) China (14.0%) United States (13.8%)	UAB Host Baltic (15.2%) DigitalOcean, LLC (13.1%) OVH SAS (3.8%)	141.98.11.30 (9.7%) 141.98.10.210 (3.5%) 179.60.147.101 (1.5%)	0.805	0.499	0.332		2.2x

Fig. 16. All columns of the Usernames tab.

TABLE I
DISCOVERY TABLE COLUMNS BY TAB

Column	Countries	IPs	ASNs	Usernames
Total attacks	✓	✓	✓	✓
Average daily	✓	✓	✓	✓
Persistence	✓	✓	✓	✓
Max absolute Δ	✓	✓	✓	✓
Max % Δ	✓	✓	✓	✓
Recent (7d)	✓	✓	✓	✓
First / last seen	✓	✓	✓	✓
Max daily	✓	✓	✓	✓
Trend	✓	✓	✓	✓
Burst intensity	✓	✓	✓	✓
Unique usernames	✓	✓	✓	
Unique IPs	✓		✓	✓
Unique ASNs	✓			✓
Unique countries				✓
Username stability	✓	✓	✓	
IP stability	✓		✓	✓
ASN stability	✓			✓
Country stability			✓	✓
Username concentration	✓	✓	✓	
IP concentration	✓		✓	✓
ASN concentration	✓			✓
Country concentration			✓	✓

Rank	IP Address	Usernames ▾	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	68.183.213.225	—	77	16	373.6K	Germany	DigitalOcean, LLC
1	143.244.183.240	100%	77	16	136.4K	United States	DigitalOcean, LLC
2	24.199.108.32	100%	77	16	140.5K	United States	DigitalOcean, LLC
3	146.190.113.228	100%	77	16	155.4K	United States	DigitalOcean, LLC
4	134.122.62.125	100%	77	16	159.1K	Netherlands	DigitalOcean, LLC
5	101.204.27.20	100%	77	16	75.5K	China	CHINA UNICOM China169 Backbone
6	116.1.214.99	99.9%	77	16	13.9K	China	CHINANET-BACKBONE
7	221.234.230.12	99.8%	77	16	122.4K	China	CHINANET-BACKBONE
8	118.184.171.58	98.1%	77	16	572	China	ChangZhou Bitcomm Software Technology Co., Limited
9	41.205.23.56	94.7%	72	16	285	Cameroon	MTN Network Solutions (Cameroon)
10	211.155.104.65	93.4%	77	16	1.4K	China	Kaopu Cloud HK Limited
11	104.248.205.76	85.8%	54	11	121.6K	Netherlands	DigitalOcean, LLC
12	68.183.221.15	84.9%	77	16	181.3K	Germany	DigitalOcean, LLC

Fig. 17. The shared-username search for `68.183.213.225` (Section V-A), showing IPs above 84% similarity across five countries and six ASNs.

Rank ▲	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	200.195.169.59	Searched	3,821	925	36.4K	Brazil	Ligga Telecomunicações S.A.
1	132.255.253.123	94.9%	1,802	46	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
2	103.186.99.250	92%	1,684	72	29.5K	Indonesia	PT Network Center Nusantara
3	79.3.19.34	90.8%	1,899	83	36.2K	Italy	Telecom Italia S.p.A.
4	201.116.12.217	89.2%	1,866	140	33.8K	Mexico	UNINET
5	27.99.9.141	88.6%	1,842	86	41.2K	Australia	Microplex PTY LTD
6	45.118.145.221	87.5%	1,525	52	26.8K	Vietnam	Long Van Soft Solution JSC
7	103.110.8.244	87.4%	1,480	57	24.9K	Indonesia	PT Citra Jelajah Informatika
8	160.120.247.113	86.9%	1,480	40	22.5K	Ivory Coast	Orange Côte d'Ivoire
9	211.46.4.196	86.9%	1,907	77	44.2K	South Korea	Korea Telecom
10	77.235.29.116	86.9%	1,526	58	25.8K	Kyrgyzstan	Mega-Line Ltd.
11	130.61.35.0	86.8%	1,554	58	27.0K	Germany	Oracle Corporation
12	202.4.42.154	86.7%	1,756	144	32.6K	Samoa	Vodafone Samoa Limited
13	43.131.43.132	86.7%	1,609	58	27.0K	Germany	Tencent Building, Kejizhongyi Avenue
14	82.160.101.186	86.7%	1,769	99	27.7K	Poland	TK Telekom sp. z o.o.
15	188.250.172.49	86.4%	1,654	49	29.8K	Portugal	MEO - SERVICOS DE COMUNICACOES E MULTIMEDIA S.A.
16	200.53.21.177	86.4%	1,889	78	39.6K	Brazil	ACESSOLINE TELECOM ACESSO (ALT)
17	202.57.17.34	86.4%	1,877	99	39.2K	Indonesia	PT. Sejuta Jaring Global
18	2.40.59.98	86.3%	1,648	47	27.9K	Italy	Vodafone Italia S.p.A.
19	81.219.237.226	86.1%	1,717	55	32.3K	Poland	Netia SA
20	188.68.31.75	86%	1,587	51	26.5K	Russia	LTD AtelRybinsk

Fig. 18. Top 20 behavioral similarity matches for `200.195.169.59` (Section V-B), spanning 14 countries and 20 ASNs.

Appendix B
Username Campaigns

This appendix extends the username campaign in Section V-A. There, the shared-username search for `68.183.213.225` (DigitalOcean, Germany) returned five IPs with 100% shared-username similarity (Fig. 9). All six IPs tried the same 77 usernames, including 16 uncommon ones. Fig. 17 shows the same search beyond those six IPs. Twelve IPs score above 84% similarity, and they span five countries and six ASNs. All but one of the 100% matches share a single ASN, but the wider set reaches other providers. This suggests that the campaign's username list is used beyond a single hosting provider, either by one operator controlling hosts on several networks or by several operators sharing a list.

Appendix C
Hidden Entities

This appendix extends the behavioral similarity search in Section V-B. There, the five closest matches to `200.195.169.59` (Ligga Telecomunicações S.A., Brazil) were each at least 88.6% similar and each came from a different country and ASN (Fig. 11). Fig. 18 shows the top 20 matches. All are at least 86% similar, and they span 14 countries and 20 ASNs, so no two matches share an ASN. No single country or ASN block would therefore catch more than a few of these IPs. The behavioral similarity search gives administrators a candidate set of IPs to consider blocking, along with the shared behavior that supports doing so.